

NEKHA BOSE

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SUMMARY

Result-driven Sr. Software Engineer with 5+ years of experience in developing scalable software solutions and leveraging data for business insights. Currently pursuing a Master's degree in Information Technology and Management at Illinois Institute of Technology. Experienced in building real-time processing systems, and visualizing data for key stakeholders. Certified Azure AI Engineer Associate, showcasing expertise in Azure AI technology and data-driven solutions.

WORK EXPERIENCE

Lead Software Developer - BreatheIT Solutions Inc

December 2024 - Present

Lifestyle Technology platform Development

- Engineered an AI powered **Lifestyle platform** with a **Node.js** backend implementing RESTful APIs with optimized query handling and responsive frontend components built using **React.js** for the business portal and **React Native** for the mobile application for both **Android and IOS**, collaborating closely with designers.
- Implemented **JWT** authentication using an access and refresh token mechanism, ensuring secure user sessions, automated token renewal, and robust access control and also built **WebSockets** for real-time notifications. Designed and managed **50+ PostgreSQL** tables across multiple databases ensuring efficient data retrieval and following best practices.
- Deployed **150+** endpoints in GCP VMs using docker and configured the endpoints using Nginx for load balancing.
- Championed code quality standards led 15+ engineers through React/**TypeScript** best practices and backend API design reviews.
- Conducted risk assessments and provided timely technical reports to stakeholders, improving process transparency. Lead **Scrum** calls, followed Agile methodologies for continuous integration and deployment, ensuring 30% faster feature rollouts.
- Performed Unit Testing and Integration testing utilizing Postman for rigorous API testing, reducing debugging cycles and ensuring seamless request-response handling. Designed and executed **A/B tests** on new UI flows in React/TypeScript, leading to a 12% lift in user engagement

Sr. Software Engineer - Infosys Limited

March 2021 - December 2023

E-commerce platform Development

- Led** the creation of a scalable E-Commerce platform using **Java (Spring Boot)** for backend and React.js for frontend, ensuring robust performance. Successfully supported over **10 Million+** concurrent users using Docker and Kubernetes, directly driving a 20% increase in user satisfaction and retention.
- Integrated MongoDB and Postgres to enhance backend data retrieval and storage, optimizing query performance and reducing response times by 40%, which significantly improved overall system efficiency and user experience.
- Conducted risk assessments and provided timely technical reports to stakeholders, improving process transparency. Participated in **Scrum and Agile**, facilitating continuous improvement and timely delivery.

Data Automation and ETL Optimization for Nike

- Spearheaded the implementation of ETL processes using **Autosys, Apache Airflow, and AWS**, reducing data processing time by 40% and improving data retrieval speed by 30%.
- Collaborated with business analysts to design and build data visualizations and reports using Tableau, providing stakeholders with clear, actionable insights, improving decision-making efficiency by **30%** and increasing team productivity by **25%**.
- Worked closely with product managers to ensure alignment between technical solutions and business objectives.

Software Engineer - Cogniz Info Tech

June 2017 - December 2019

Interactive Dashboard for Sales Analytics

- Developed an interactive dashboard using React.js for sales analytics, improving user engagement by 20% through optimized typography, enhancing readability and user interaction.
- Implemented **GraphQL** APIs in our Node.js/Express backend to streamline data fetching for the React dashboard, reducing over-fetching by 30%.
- Reduced backend error rate by 35% via enhanced validation and retry logic.
- Integrated the frontend with a Node.js/Express backend to fetch and process data from a **MongoDB** database. Utilized RESTful APIs to ensure seamless data flow, providing users with up-to-date sales insights and performance metrics, resulting in a 30% improvement in sales decision-making efficiency.

PROJECTS

Travel Tracker: MVVM Android App with Firebase, Gemini AI & Interactive Map

January 2025 - May 2025

- Employed **MVVM** with Hilt for dependency injection, LiveData & ViewModel, and **Kotlin** Coroutines to integrate Firebase Authentication (Email/Password) and Realtime Database, using JSON serialization for efficient, private trip data handling.
- Leveraged Google AI SDK (Gemini) via RESTful API integration for dynamic itinerary generation, and used Jetpack Lifecycle Components alongside Geospatial Coordinates & Geocoding APIs to display trips on Google Maps with accurate markers

SmartMart – Full-Stack ECommerce Web App

September 2024 – December 2024

- Built a role-based multi-user e-commerce platform using Jakarta EE and JSF with Facelets, supporting customer, seller, and admin workflows with Jakarta Security-driven authentication and authorization. Architected complex entity relationships with ORM (JPA) and enforced transactional integrity for User, Product, Order, and OrderLine entities.
- Created reusable composite UI components, integrated Bootstrap for responsive design, and enabled dynamic routing with JSF navigation

flows.Utilized MVC design pattern, DAO layers, and modular code organization to ensure maintainability and scalability and configured custom security realms and data sources for deployment on **Payara Server**.

SKILLS

- **Front-end:** React.js, Redux, React Native, Angular, HTML5, CSS3, SCSS, JavaScript, TypeScript, jQuery, D3.js, FlutterFlow, Bootstrap, web development
- **Back-end & APIs:** Node.js/Express, Python, Java (Spring Boot), REST APIs, GraphQL, Maven, Microservices, Serverless (AWS Lambda), JavaFX
- **DevOps:** Docker, containerization, Kubernetes, CI/CD (Jenkins, GitHub Actions), Nginx, PM2
- **Data & Cloud Technologies:** ETL processes with Autosys, Apache Airflow, AWS Glue, Data transformations in Teradata, Tableau, Real-time Data Processing, Model Deployment, AWS (EC2, Lambda), GCP (Compute Engine, App Engine, Cloud Run), Kafka.
- **Databases:** SQL(Postgres, SQLite), NoSQL(MongoDB), Amazon S3, Google Cloud Storage
- **Testing & Tools:** A/B Testing,Jasmine, JUnit, Postman, Swagger, Webpack, Babel
- **Methodologies:**Scrum, Agile, Kanban, WaterFall, SaaS, B2B, B2C, SDLC, Feature Grooming, Story Point Estimation, Code Reviews.
- **Design & Collaboration:** Figma, Sketch, Jira, Confluence, InVision, Wireframe Translation, UI/UX Implementation

EDUCATION

Illinois Institute of Technology, Chicago, IL

Information Technology and Management - GPA 3.88/4.0

APJ Abdul Kalam Technological University, Kerala

Information Technology

ACHIEVEMENTS

- **Microsoft Certified: Azure AI Engineer Associate** from Microsoft, demonstrating expertise in Azure AI technology.
- Awarded the Best Student Project (Braille Converter) in the University for the year 2019-20.
- Presented a paper in the SERB sponsored National Conference on recent Developments in Computational Method for renewable energy.